

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

SSC PART-I (CLASS 9) PRACTICE EXAMINATION

CHEMISTRY — PAPER 23 — ANSWER KEY (Version A)

Syllabus: Chapter 10 – Acids, Bases and Salts (35 marks) + Revision: Chapters 4, 6, 8, 9 (15 marks)

Total Marks: 50

New chapter: 35

Revision: 15

SECTION A — ANSWER KEY (10 × 1 = 10 MARKS)

1. In $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$, water is behaving as a — (c) Bronsted acid 1

Water **donates a proton** to the ammonia molecule and is left as OH^- . A Bronsted-Lowry acid is a **proton donor**, so in this particular reaction water is the acid and NH_3 is the base. Water can act as an acid in one reaction and a base in another, which is why it is called **amphoteric**.

Source: Ch 10, section 10.1.2, book p143 (" H_2O is converted to OH^- by donating a proton, so H_2O is an acid"). SLO: define Bronsted-Lowry acids as proton donors and bases as proton acceptors.

2. A base but not an alkali — (b) $\text{Cu}(\text{OH})_2$ 1

An **alkali is a base that dissolves in water**. Copper hydroxide is a hydroxide of a metal, so it is a base, but it is **insoluble in water**, so it is not an alkali. NaOH , KOH and $\text{Ca}(\text{OH})_2$ all dissolve, so all three are alkalis.

Source: Ch 10, "Alkalis", book p145. SLO: recognize that bases are oxides or hydroxides of metals and that alkalis are water-soluble bases.

3. A strong acid — (a) It dissociates completely into ions in aqueous solution 1

Strength is about **how completely the acid ionizes**, not about how much of it dissolves and not about how many hydrogen atoms the formula shows. HCl , HNO_3 and H_2SO_4 ionize 100% in water, so they are strong. Option (d) is a property of a base, not an acid.

Source: Ch 10, section 10.2.1, book p144. SLO: define a strong acid as one that completely dissociates in aqueous solution.

4. Gas from marble + dilute sulphuric acid — (d) carbon dioxide 1

Marble is calcium carbonate. **Acid + metal carbonate → salt + water + carbon dioxide**. Hydrogen is the trap here: hydrogen comes from **acid + metal**, not from acid + carbonate.

Source: Ch 10, section 10.3 point 4, book p145 ($\text{CaCO}_3 + \text{H}_2\text{SO}_4$). SLO: describe the characteristic properties of acids in terms of their reactions with metals, bases and carbonates.

5. Gas from ammonium chloride + sodium hydroxide — (a) ammonia 1

Bases decompose ammonium salts on heating: **base + ammonium salt → salt + ammonia + water**. $\text{NaOH} + \text{NH}_4\text{Cl} \rightarrow \text{NaCl} + \text{NH}_3 + \text{H}_2\text{O}$.

Source: Ch 10, section 10.4 point 5, book p146. SLO: identify the characteristic properties of bases in terms of their reactions with acids and ammonium salts.

6. Every acid solution contains an excess of — (b) H^+ ions 1

An acid ionizes in water to give H^+ ions; an alkali ionizes to give OH^- ions. Getting this pair the right way round is the whole basis of the chapter.

Source: Ch 10, section 10.1.1, book p141. SLO: recognize that aqueous solutions of acids contain H^+ ions and aqueous solutions of alkalis contain OH^- ions.

7. Ordinary rain is slightly acidic because of dissolved — (c) carbon dioxide 1

The book is precise about this: **normal rain is slightly acidic because of dissolved carbon dioxide**. SO_2 and NO_2 are the acid oxides that come from human activity and turn normal rain into **acid rain**, which is a different thing.

Source: Ch 10, end of section 10.3, book p145. SLO: define acid rain.

8. [REVISION, Ch 8] Catalyst for hydrogenating vegetable oils into margarine — (d) nickel 1

Nickel is the catalyst for the manufacture of margarine. All four options are on the same list in her book, so learn them as one set: **iron** → Haber Process (synthesis of ammonia); **platinum and palladium** → catalytic converters in automobiles; **nickel** → manufacture of margarine.

Source: Ch 4, section 4.5 “Transition Elements”, point 5 “Catalysts for Industrial Processes”, book p64. Re-test: answered “platinum” on exam-22 Q6, and “nickel” for the Haber process on exam-21 Q14. This is the second re-test of this list.

9. [REVISION, Ch 4] Element X with configuration 2, 8, 2 — (a) Group 2, Period 3 1

Period = number of occupied shells = 3. **Group** for an **s-block** element = number of valence electrons = 2. The **+10 correction belongs only to p-block elements**, which is exactly the slip made on exam-22, where 2 + 10 was written for an s-block element.

Re-test: exam-22 Section B, element X ($3s^2$) answered as Group 8 after adding then striking out 10. Ch 4 SLO: electronic configuration and position in the periodic table.

10. [REVISION, Ch 9] At dynamic equilibrium the forward rate is — (b) equal to the rate of the reverse reaction 1

At equilibrium the two rates are **equal**, not zero and not greater. Both reactions are still happening, which is why it is called **dynamic**; the amounts stop changing because the two rates cancel out.

Re-test: this exact pair (rates equal vs greater) was missed on the OPF First Term school paper, August 2026. Ch 9 SLO: dynamic equilibrium.

SECTION B — ANSWER KEY ($6 \times 4 = 24$ MARKS)

2. (i) Define a Bronsted-Lowry acid and a Bronsted-Lowry base. 2

A Bronsted-Lowry **acid** is a **proton donor** (it gives away an H^+ ion). 1

A Bronsted-Lowry **base** is a **proton acceptor** (it takes an H^+ ion). 1

(ii) $HCl + H_2O \rightarrow H_3O^+ + Cl^-$ — identify the acid and the base, with a reason. 2

HCl is the Bronsted-Lowry acid, because it donates a proton and is converted into Cl^- . 1

H_2O is the Bronsted-Lowry base, because it accepts that proton and becomes the hydronium ion H_3O^+ . 1

Useful check from the book: after donating a proton an acid forms a negative ion, and after accepting one a base forms a positive ion.

Source: Ch 10, section 10.1.2 and Example 10.1, book p142–143.

3. Ionization equations, with the correct arrow in each case. 4

(i) $HNO_3(l) \rightarrow H^+(aq) + NO_3^-(aq)$ **single arrow** 1

(ii) $H_2SO_4(l) \rightarrow 2H^+(aq) + SO_4^{2-}(aq)$ **single arrow, and note the 2 in front of H^+** 1

(iii) $CH_3COOH \rightleftharpoons H^+ + CH_3COO^-$ **reversible arrow** 1

(iv) **CH_3COOH (ethanoic acid) is the weak acid**, because it does not ionize completely in water. Only about 5% of its molecules ionize, so a reversible arrow must be used. 1

Marking note: the reversible arrow on (iii) and single arrows on (i) and (ii) are the marks here. A single arrow on ethanoic acid loses the mark even if the products are right. Source: Ch 10, sections 10.2.1 and 10.1.1, book p141 and p144.

4. Balanced equations with state symbols, and the gas named. 4

(i) $Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$ 1 balanced ½ state symbols

Gas released: **hydrogen** ½

(ii) $Na_2CO_3(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$ 1 balanced ½ state symbols

Gas released: **carbon dioxide** ½

Marking note: the book prints both of these unbalanced ($Na_2CO_3 + HCl$), so the 2 in front of HCl and the 2 in front of $NaCl$ are genuinely her own work. State symbols were the instruction she missed on exam-22; the paper warns about them in instruction 6, so the half mark is deducted if they are absent. Source: Ch 10, section 10.3 points 3 and 4, book p145.

5. (i) All alkalis are bases, but all bases are not alkalis. Explain. 2

An **alkali is a base that is soluble in water.** 1 Every alkali is therefore a base, but many bases do not dissolve in water, and those insoluble bases are **not** alkalis. So alkalis are a smaller group inside the bases. 1

(ii) One base that is an alkali, one that is not, with formulae. 2

Alkali: **sodium hydroxide, NaOH** (or KOH, or Ca(OH)₂) 1

Not an alkali: **copper hydroxide, Cu(OH)₂** (or Al(OH)₃, or Fe(OH)₃) 1

Source: Ch 10, "Alkalis" paragraph, book p145.

6. (i) Two characteristic properties of bases, other than taste. 2

Any **two** of the following, 1 mark each:

- They change the colour of **red litmus paper to blue.**
- Their aqueous solutions feel **slippery** to the touch.
- They **neutralize acids** to form a salt and water.
- They **decompose ammonium salts on heating** and liberate ammonia gas.
- Their aqueous solutions **conduct electricity.**

(ii) KOH with NH₄Cl, and the gas named. 2



Gas liberated: **ammonia (NH₃)** 1

Source: Ch 10, section 10.4, book p146. The general pattern is base + ammonium salt → salt + ammonia + water.

7. [REVISION, Ch 6] 0.2 mol of NaOH. 4

(i) Molar mass of NaOH = 23 + 16 + 1 = **40 g mol⁻¹** 1

(ii) mass = number of moles × molar mass = 0.2 × 40 = **8 g** 2 (1 for the correct formula, 1 for the correct answer with the unit)

(iii) formula units = 0.2 × 6.02 × 10²³ = **1.204 × 10²³ formula units** 1

Marking note: she must finish with a sentence that answers the question, for example "She must weigh out 8 g of sodium hydroxide."

The missing-conclusion habit cost marks on six papers before it was broken on exam-21, and this question is here to keep it broken. Ch 6 SLO: the mole, molar mass and Avogadro's number.

8. Acid rain. 8**(i) Define acid rain.** 2

Normal rain is **already slightly acidic** because carbon dioxide from the air dissolves in it. 1 As a result of human activity, many **acid oxides enter the atmosphere**, and these make the rainwater **much more acidic**. Rain of this kind is called **acid rain**. 1

(ii) The two acids dissolved in acid rain. 2

Nitric acid, HNO_3 1 and sulphuric acid, H_2SO_4 1

(iii) Sulphuric acid with calcium carbonate. 2

Products: **calcium sulphate, water and carbon dioxide** 1 (*all three names needed for the mark*)

(iv) Two further kinds of damage, tied to the properties of acids. 2

Any **two**, 1 mark each. The mark is for linking the damage to a property of acids, not just for naming the damage:

- **Acids corrode metals.** Acid rain therefore attacks metal structures, bridges, railings and the metal parts of buildings, forming a salt and releasing hydrogen gas.
- **Acids react with metal carbonates.** Buildings and statues built of limestone or marble are eaten away, because the calcium carbonate in them is converted into a salt, water and carbon dioxide.
- **Acids are corrosive.** Acid rain damages the surfaces of buildings and statues it falls on.

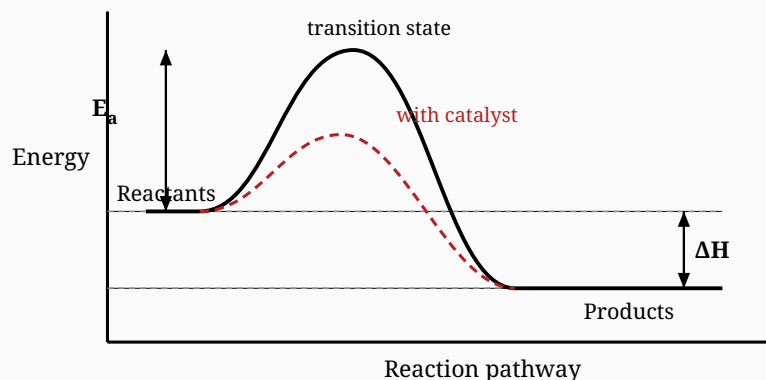
Source: Ch 10, end of section 10.3 and start of section 10.4, book p145–146 (“Due to the corrosive nature of acids, acid rain can damage structures, buildings, and statues containing metals and metal carbonates”). SLOs: define acid rain; discuss effects of acid rain and relate them with the properties of acids.

9. [REVISION, Ch 8] Energy changes in a reaction. 8

(i) Define activation energy. 2

Her book's wording (section 8.5): activation energy is **the minimum amount of energy that, in addition to the average kinetic energy, particles must have 1 in an effective collision. 1** Accept any answer carrying both ideas: a minimum energy, and that it is needed for an effective (successful) collision. The book adds that no reaction occurs if the energy of the reacting particles is lower than the activation energy.

(ii) Labelled energy-profile diagram for an exothermic reaction. 4



Solid black curve = the reaction without a catalyst. **Red dashed curve** = the same reaction with a catalyst.

The products sit **lower** than the reactants, which is what makes the reaction exothermic, so ΔH is **negative**.

Marks for the diagram: energy on the vertical axis and reaction pathway on the horizontal axis 1; products drawn **below** the reactants with both labelled 1; the hump drawn and E_a labelled from the reactants up to the top of the hump 1; ΔH labelled between the reactant and product levels, marked as negative 1.

(iii) The catalysed pathway. 2

The catalysed curve is drawn with a **lower hump** starting and ending at the same two levels. 1

E_a is **lowered** by the catalyst, but ΔH is **completely unchanged**, because the reactants and the products are still at the same energies as before. 1

The model for this diagram is Fig. 8.1 in her own book, "Energy level diagram for exothermic reaction" (p130), which carries exactly these labels: transition state, activation energy, reactants, enthalpy change $\Delta H < 0$, products, reaction progress. The catalysed curve is Fig. 8.3 (p131). Marking note: on exam-22 this diagram was drawn with no E_a label and no catalysed curve, so both are explicitly demanded in the wording this time. The "catalyst does not change ΔH " point was fixed on exam-22 and is being held here. Ch 8 SLOs: activation energy, enthalpy change, and the effect of a catalyst.

Mark distribution check

Section	Chapter 10 (new)	Revision (Ch 4, 6, 8, 9)	Total
A — MCQs	7 (Q1–Q7)	3 (Q8, Q9, Q10)	10
B — Short	20 (Q2–Q6)	4 (Q7)	24
C — Long	8 (Q8)	8 (Q9)	16
Total	35 (70%)	15 (30%)	50

Answer letter balance for Section A: a = 3 (Q3, Q5, Q9), b = 3 (Q2, Q6, Q10), c = 2 (Q1, Q7), d = 2 (Q4, Q8).